

JINGCHUN XIE

Postdoctoral Fellow

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PROFILE

My current study focuses on the interactions between mantle dynamics and surface tectonic processes, with the goal of revealing the thermodynamic histories of terrestrial planets.

I am committed to integrating a variety of research methods—including numerical simulation, geophysical inversion, geologic principals, and geostatistics—to provide a systematic view of planetary evolution paths. I earned my Ph.D. in Physics (Planetary Science) in June 2023 through the joint doctoral program of Shanghai Astronomical Observatory (Chinese Academy of Sciences) and ShanghaiTech University. During my Ph.D., I investigated the thermodynamic evolution of Mercury and its lithosphere as recorded by thrust fault-related landforms, primarily through numerical simulations.

EDUCATION

Shanghai Astronomical Observatory, Chinese Academy of Sciences & ShanghaiTech University
Shanghai, China Sep 2017 – Jun 2023

Ph.D. in Physics (Planetary Science), Joint Doctoral Program

Advisor: Chengli Huang

Dissertation: *The study of thermodynamic evolution of Mercury's lithosphere by shortening tectonic landforms*

Chengdu University of Technology, Chengdu, China Sep 2012 – Jun 2016

B.Eng. in Spatial Information and Digital Technology (GIS), College of Geophysics

Thesis: *Comparison of application in Nepal Earthquake Landslides evaluation based on SVM and Random Forests*

PROFESSIONAL APPOINTMENTS

The Chinese University of Hong Kong, Hong Kong SAR, China Oct 2023 – Present
Postdoctoral Fellow, Department of Earth and Environmental Sciences

PUBLICATIONS

Peer-Reviewed Publications

- [1] **Xie, J.-C.**, Y. Zhan, et al. (2026). Waned mantle upwelling contributes to a heavily deformed northern smooth plains on Mercury. *Nature Communications*, 17, 9064.
- [2] **Xie, J.-C.**, and C.-L. Huang (2023). On the formation of thrust fault-related landforms on Mercury: The key parameters controlling the mechanical structure of the lithosphere. *Icarus*, 401, 115594.
- [3] **Xie, J.-C.**, C.-L. Huang, and M. Zhang (2022). On the formation of thrust fault-related landforms in Mercury's Northern Smooth Plains: A new mechanical model of the lithosphere. *Icarus*, 388, 115197.
- [4] **Xie, J.-C.**, M. Zhang, and C.-L. Huang (2022). Influence of megaregolith on the thermal evolution of Mercury's silicate shell. *Research in Astronomy and Astrophysics*, 22, 035026.

Manuscripts under Review

- Jiang, Y.-R., J.-Q. Li, L. Pan, **Xie, J.-C.**, et al. Foundation models trained on real-world images enable planetary-scale geological mapping on Mars. *Submitted to National Science Review*.

Manuscripts in Preparation

- **Xie, J.-C.**, Y. Zhan, et al. Spatially heterogeneous lithospheric thermal conditions on Mercury revealed by flexural modeling and gravity–topography inversion. *In preparation*.
- **Xie, J.-C.**, Y. Zhan, et al. Estimated seismicity of Mercurian thrust fault-related landforms based on underlying geometries inversion. *In preparation*.

GRANT PARTICIPATION

- *Quantitative Analysis and Numerical Modeling of Planetary Structures: Insights into the Tectonic Evolution of Terrestrial Worlds*. 1+1+1 CUHK–CUHK(SZ)–GDSTC Joint Collaboration Fund, Award No. 2025A0505000079. PI: Yan Zhan (Assistant Professor). Role: Main Participant.
- *The Rotation and Deep Interior Structure of Terrestrial Planets*. National Natural Science Foundation of China (NSFC), Award No. 11973072. PI: Mian Zhang (Associate Professor). Role: Main Participant.
- *Study on the Driving Mechanism of the Lunar Dynamo Magnetic Field: Viscous Dissipation at the Core–Mantle Boundary Induced by Precession*. National Natural Science Foundation of China (NSFC), Award No. 12173068. PI: Shengxia Gong (Associate Professor). Role: Participant.

HONORS AND AWARDS

- Graduate Scholarship, ShanghaiTech University 2018 – 2022

CONFERENCE ACTIVITY

- The Greater Bay Earth & Environmental Sciences Young Scholars Symposium, Hong Kong SAR, China
Dec 2025
- CAS Key Laboratory of Planetary Science & State Key Laboratory of Lunar and Planetary Science Joint Annual Academic Meeting, Tongren, China
Nov 2025
- CAS Key Laboratory of Planetary Science & State Key Laboratory of Lunar and Planetary Science Joint Annual Academic Meeting, Xiamen, China
Dec 2024
- The SYSU–CUHK Joint Symposium, Zhuhai, China
Nov 2024
- The 4th Structural Geology and Geodynamics Forum, Zhuhai, China
Mar 2024
- The Greater Bay Earth & Environmental Sciences Young Scholars Symposium, Hong Kong SAR, China
Nov 2023
- The 5th Young Scientist Forum of Planetary Science, Sanya, China
Mar 2023
- CAS Key Laboratory of Planetary Science & State Key Laboratory of Lunar and Planetary Science Joint Annual Academic Meeting, Chengdu, China
Nov 2019